

Labwork 6.1 - Find Word Pattern

General Information and Submission Rules:

- By the end of the labwork, you must submit it to Yulearn as **studentID_fullname_labwork6.c**.
- Incorrect naming or file format will result in a loss of 5 points from your total grade.
- Only use concepts covered in the labs so far.

Question 1 - Display Message (25p)

- Create a new function (**welcome**) that displays a welcome message and a list of all available operations to the user.
- This function will expect a decision from a user so that it will execute related functions based on this choice until the user exits.
- Make sure to print each line separately.

```
Welcome to Labwork 6 Machine! By using this machine you can do:
1. Array Summation
2. Divisible by 5
3. Word Search
4. Exit
Make your choice:
```

Question 2 - Array Summation (25p)

- Create a new function (**arr_sum**) that takes two arrays and their sizes as parameters. The size needs to be determined by the user.
Input: array_1, array_2, arr_size
Output: print result_arr
- Firstly, the function should compare pairwise elements of two arrays. If the element in the first array is greater than the corresponding element in the second array at position i, then the integer result array's i-th position should be their summation; otherwise, it should be the second array's i-th value.

```
>> Enter the size: 5
>> Enter the values of array 1: 1 8 9 12 56
>> Enter the values of array 2: 9 4 2 86 2
The result array is: 9 12 11 86 58
```

Question 3 - Divisible by 5 (25p)

- Create a new function (**divisible_three**) that takes the array as a parameter and in the end this function will return the values that are divisible by 5. You cannot use the modulus operator (%) assuming each number is 2 digits.

```
>> Enter the size: 5
>> Enter the values of array: 15 10 11 86 58
The values: 15 10
```

Question 4 - Word Search (25p)

- Create a new function (**word_search**) that tries to find out the given word inside the given string. You can directly add given sentence and word to your function like this:

```
char sentence[] = "a mischievous squirrel cleverly hid its biggest acorn beneath a
pile of freshly fallen maple leaves"; //size 99
char word[] = "acorn"; //size 5
```

- In the end this function will display whether the given word is inside the sentence.

```
Found!
```